Event-Driven Robot Coding

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March 20, 2026

Packet Use

This printable lesson packet is generated from the paired Markdown guide. The Markdown version remains the clickable, neurodivergent-friendly working copy; this PDF carries the same lesson substance in a formal handout format. Evidence claims in this packet are based on transcript-derived lesson notes and the cleaned transcript files in this workspace.

Quick Scan

- **Grade:** Grade 4
- **Grade Hub:** Notes [../grade-4-Notes.md]
- **Schedule Reference:** Spring2026_TeachingSchedule_Derek.md [../../school/Spring2026_TeachingSchedule_I
- **Workbook Sheet:** `4th` in `Teaching_Placement-IGNITE Curriculum_ GCSD25-26.xlsx`
- **Lesson Focus:** Students use Sphero sensor or event blocks to trigger robot behavior, then debug how input, environment, and code interact.
- **CS Alignment Strength:** Strong CS lesson
- **LaTeX Source:** event-driven-robot-coding.tex [./event-driven-robot-coding.tex]
- **Bib File:** event-driven-robot-coding.bib [./event-driven-robot-coding.bib]
- **Artifacts Folder:** artifacts/ [./artifacts/]

Lesson Identity

- **Likely Class Blocks:** Day 1 – 1:15-2:10 – Baris, Day 2 – 1:15-2:10 – Fowler, Day 3 – 1:15-2:10 – Schrank, Day 4 – 1:15-2:10 – Autore
- **Main Lesson Family:** Event-Driven Robot Coding
- **Primary Evidence Base:** Transcript-derived notes plus source transcripts from this teaching-placement workspace.
- **Primary External Source Type:** Official curriculum/provider materials.

What We Are Teaching

- what an event is in block coding
- how sensor input changes program behavior
- how to predict trigger-result relationships
- how to test and debug event-driven code

CS Standards Alignment

- **1B-AP-10**: Create programs that include sequences, events, loops, and conditionals. Why it fits here: Students explicitly build around event blocks and sensor-triggered behavior.
- **1B-AP-11**: Decompose problems into smaller, manageable subproblems. Why it fits here: Students isolate trigger, action, and environment as separate pieces of the task.
- **1B-AP-15**: Test and debug a program or algorithm to ensure it runs as intended. Why it fits here: Students must revise code and setup when the robot does not react correctly.

NYS / Local Curriculum Alignment

- Aligned to the local Grade 4 robotics progression in the workbook and schedule.
- Supports New York-facing technology goals through explicit computational problem solving, testing, and revision.
- This lesson is a strong CS lesson because event-driven logic is the core content, not a side feature.

Lesson Content and Concepts

- event
- sensor
- input
- output
- debugging

Evidence / Proof of Instruction

- **Transcript-derived note:** 260303 Grade 4 Baris Event Sensors [../../transcript-derived-lessons/260303-grade-4-baris-event-sensors.md] The note directly names ambient light sensors and flashlight-triggered coding.
- **Transcript-derived note:** 260312 Grade 4 Baris Agent Challenge [../../transcript-derived-lessons/260312-grade-4-baris-agent-challenge.md] Later challenge work shows students applying similar event/debug logic in a more complex environment.

- **Source transcript:** 260303 Grade 4 Baris Event Sensors [../../transcripts/260303-grade-4-baris-event-sensors.md] The source transcript records discussion of events, sensors, and New York-required learning in this kind of class.
- **Likely student proof:** Working event-triggered programs, failed attempts with documented fixes, and sensor-response demonstrations all count as proof.

Assessment Evidence

- Student can explain what event causes the robot action.
- Student can build or modify an event-driven block stack.
- Student can identify whether failure came from code, environment, or sensor setup.

UDL and Inclusion Supports

Multiple Representation

- Demonstrate the event with a visible physical trigger before showing the block code.
- Keep trigger and response language paired on a chart: when this happens, do this.
- Use one predictable test area for all students first.

Multiple Action / Expression

- Allow students to predict orally, with arrows, or in block form before launch.
- Use pair roles such as flashlight/controller and coder.
- Provide a debug checklist with three questions: trigger, code, environment.

Multiple Engagement

- Use immediate robot response as motivation for repeated testing.
- Frame unexpected behavior as a clue rather than failure.
- Keep challenges short enough for several iteration cycles.

How We Are Already Making It Inclusive

- The transcript evidence shows the class was already being taught as explicit CS, not just robot play.
- Physical triggers and sensor-based tasks give concrete access to abstract event language.
- Teacher framing connected the work to required learning, which strengthens legitimacy and purpose.

How to Improve Inclusion Next Time

- Add event-response anchor cards at every table.
- Prepare one low-floor sensor challenge and one stretch challenge.
- Capture short video clips of working triggers in `artifacts/` for proof and reteaching.
- Add a reflection prompt: what changed when the environment changed?

Materials and Setup

- Sphero robots with sensor features
- Student devices
- Flashlights or trigger objects
- Open test space
- Debug checklist

Teacher Moves / Script / Routines

- Physically demonstrate the event before naming the block.
- Ask students to predict first, then launch.
- Stop to compare one working event stack and one non-working stack.
- Close by having students name the trigger and the robot response.

Common Errors and Debugging

- Students may confuse repeated motion with event-driven behavior.
- Sensor placement or lighting conditions can cause inconsistent results.
- Students may change too many things at once while debugging.

Extensions and Simplifications

- Extension: compare two different event triggers for the same robot response.
- Simplification: use one event and one response only.
- Extension: add a condition or loop after the event logic works.

Related Local Materials

- No direct local lesson packet linked yet.

Artifacts Checklist

- Compiled lesson PDF in `artifacts/`
- At least one screenshot, student example, or setup photo
- One short assessment or observation record
- Updated standards/source bibliography once the `.bib` file is revised
- Any teacher reflection or revision notes after reteaching

Selected Official Sources

Selected official sources that informed this packet are listed below. ocite*

References